

Proposed Solution

Date	17 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API
Maximum Marks	2 Marks

Proposed Solution

Element	Description
Project Title	AI FAQ Assistant – AI-Powered FAQ & Customer Support Automation Backend
Objective	Build a secure, RESTful Node.js/Express backend that lets an authenticated creator manage categorized FAQ entries, exposes keyword search for anyone, and uses Google Gemini to answer support questions directly or draft new FAQ pairs from a topic.
Scope	User authentication (JWT + bcrypt), full CRUD and keyword search on FAQ records scoped by category and creator, and two Gemini-backed AI endpoints (direct answer, FAQ-pair generation). The API is validated through Postman/Thunder Client using the team's own collection.
Problem it Solves	Removes the manual burden of drafting and organizing FAQ content, replaces ad-hoc document search with structured keyword search, and adds an authorship check so only the creator of an entry can change it.
Proposed Solution / Approach	A modular Express application (controllers / middleware / models / routes / services / utils under src/) backed by MongoDB via Mongoose, JWT-protected write routes, and a dedicated services/geminiService.js that isolates all Gemini calls from the rest of the business logic.
Resource Requirements	Node.js v18+, npm v9+, Express.js, MongoDB, Mongoose, jsonwebtoken , bcrypt , Google Gemini API key (@google/genai), VS Code + Postman/Thunder Client, Git & GitHub, 5-member team over the 15–29 September 2026 cycle.